



Kacti Gen AI – Bootcamp Syllabus

1. Introduction to GenAI Engineering

Start your journey into GenAI engineering. Understand the key concepts and tools used in the industry. Gain a comprehensive overview of AI and machine learning technologies. Set the stage for advanced learning and practical application. Lay a strong foundation for your GenAI career.

2. Large Language Models (LLM) Fundamentals

Dive deep into Large Language Models including OpenAI GPT series, Claude (Anthropic), Gemini (Google), and Llama Family (Meta). Understand model architectures, training processes, and capabilities. Learn prompt engineering, fine-tuning techniques, and model selection strategies for different use cases.

3. LangChain Ecosystem Mastery

Master the complete LangChain ecosystem including LangChain core framework, LangGraph for graph-based workflows, LangFuse for observability and analytics, and LangSmith for development and debugging. Build complex AI applications with chain-of-thought reasoning and multi-step workflows.

4. Vector Databases and Embeddings

Master vector databases including Qdrant, Weaviate, FAISS, Milvus, and Chroma for AI-native applications. Learn embedding techniques, similarity search, and retrieval-augmented generation (RAG). Implement semantic search, document retrieval, and knowledge base systems for GenAI applications.

5. Model Context Protocol (MCP) and Agent-to-Agent Communication

Learn Model Context Protocol for standardized AI model communication and Agent-to-Agent (A2A) communication patterns. Implement multi-agent systems, orchestrate complex workflows, and build collaborative AI applications. Master inter-agent messaging, task delegation, and distributed AI processing.

6. Local LLM Deployment with Ollama

Master local LLM deployment using Ollama for privacy-focused and cost-effective AI applications. Learn model quantization, optimization techniques, and hardware requirements. Implement offline AI capabilities, custom model fine-tuning, and edge deployment strategies for production environments.

7. HuggingFace Ecosystem and Model Hub

Master the HuggingFace ecosystem for model deployment, sharing, and collaboration. Learn to use Transformers library, Datasets, and Spaces for building and deploying AI applications. Implement model fine-tuning, dataset preparation, and community collaboration through the HuggingFace platform.

8. Database Technologies for GenAI Applications

Master multiple database technologies including MongoDB, PostgreSQL with SQLAlchemy and Alembic, Supabase, and SQLiteDB. Learn to design data architectures for AI applications, implement database migrations, and optimize queries for large-scale GenAI systems with proper data modelling.

9. Workflow Orchestration and Background Processing

Master workflow orchestration using Prefect and background job processing with Dramatiq for scalable GenAI applications. Learn to design complex AI pipelines, handle asynchronous tasks, and implement distributed processing systems for production-grade AI workflows.

10. Interactive Development with Notebook LLM and PyNotes

Master interactive development environments using Notebook LLM for AI experimentation and PyNotes for technical documentation. Learn to create reproducible research notebooks, document AI experiments, and build interactive AI prototypes using modern notebook-based development workflows.

11. Alternative GenAI Frameworks and Haystack

Explore alternative GenAI frameworks beyond LangChain, focusing on Haystack NLP pipeline framework. Learn to build production-ready NLP applications, implement custom components, and compare different framework approaches for various GenAI use cases and requirements.

12. Research Publication and Portfolio Development

Complete your GenAI journey by publishing 1-3 research papers on ArXiv and building a comprehensive portfolio. Learn academic writing, research methodologies, and publication processes. Create projects for GitHub, GitBook documentation, Google Colab notebooks, and HuggingFace model sharing to showcase your expertise.

Duration of the Course: 4 Months

Fees: 2500 per month (Original Fees – 95,000)

Presence: India, Canada, Australia, USA, UK, UAE, Germany, Netherlands, France

Prerequisites: Basic Python Knowledge

Contact: